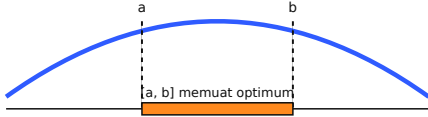


Cheat Sheet Materi 6

TI5001 / Optimisasi & Intelegensia Komputasional · Metode Numerik Single

1 · Konsep dasar



- **Unimodal**: fungsi punya satu puncak atau satu lembah (asumsi metode numerik)
- **Interval ketidakpastian**: interval tempat optimum berada; dipersempit sampai panjang akhir **1**
- Bila tidak unimodal (mis. dua lembah) → sebagian interval bisa dibuang salah → hanya optimal lokal

2 · Peta & perbandingan metode

Metode	Basis	Input	f baru
Direct search tak berpembatas	nilai f	$x^{(1)}, l$	1
Newton tak berpembatas	f', f''	$x^{(1)}, \epsilon$	-
Dichotomous $a \leq x \leq b$	nilai f	$[a, b], \epsilon, l$	2
Fibonacci $a \leq x \leq b$	nilai f	$[a, b], \epsilon, l$	1
Golden section $a \leq x \leq b$	nilai f	$[a, b], l$	1
Bisection $a \leq x \leq b$	f'	$[a, b], \epsilon$	-

Direct search & Newton butuh titik awal; metode lain butuh interval $[a, b]$.

3 · Direct search

- Langkah $\lambda = l/2$; arah $d=+1$ (kanan) / -1 (kiri)
- Mulai $x^{(1)}$: bila $f(x+\lambda d)$ membaik → maju; jika memburuk → balik $d=-1$; jika sama → berhenti
- $x^{(k+1)} = x^{(k)} + \lambda d$ selama membaik; berhenti saat $f(x+\lambda d)$ tak lebih baik

Slide: $\min (x-1)^2, l=0,01 (\lambda=0,005), x^{(1)}=3: f(3,005)=4,020 > f(3)=4 \rightarrow d=-1; 401$ iterasi; berhenti di $x=1,000 \rightarrow$ interval **[0,995; 1,005]** (panjang $2\lambda=l$)

† $(x-2)^2, x^{(1)}=5, \lambda=0,05: x=5-0,05k$; berhenti di $x=2,00 (k=60)$, interval $[1,95; 2,05]$.

4 · Newton (optimisasi)

Pendekatan kuadratik di $x^{(k)}$:

$$q(x) = f + f'(x-x^{(k)}) + \frac{1}{2}f''(x-x^{(k)})^2$$

$$x^{(k+1)} = x^{(k)} - f'(x^{(k)}) / f''(x^{(k)})$$

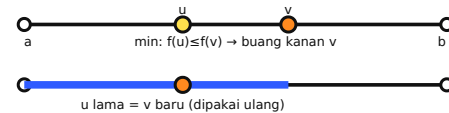
dari $q'(x)=0$. Henti: $|x^{(k+1)} - x^{(k)}| < \epsilon$ atau $|f'| < \epsilon$

Syarat: f dapat diturunkan 2x, titik awal; $f''(x^{(k)})=0 \rightarrow$ pembagian nol (mis. x^4-2x di $x=0$)

Slide: $\min (x-1)^2, f'=2(x-1), f''=2, x^{(1)}=0: x^{(2)}=0-(-2)/2=1; x^{(3)}=1 \rightarrow$ stop
† $3x^2-6x+2$ dari 10: $10-54/6=1$ (kuadrat: 1 langkah). x^3 di $x=1: f=1, f'=3, f''=6 \rightarrow x=1-3/6=0,5$

Newton optimisasi: f'/f'' . Newton-Raphson akar: f/f' .

5 · Dichotomous search



$$u = a + (b-a)/2 - \epsilon/2; v = a + (b-a)/2 + \epsilon/2$$

ϵ = konstanta pembeda; henti jika $b-a < l$; 2 evaluasi baru per iterasi

Slide: $\min (x-1)^2, [-1,2], \epsilon=0,0001, l=0,01$

k	u	v	f(u)	f(v)
1	0,5	0,5001	0,25005	0,24995
2	1,2499	1,25	0,062463	0,062513

$k=1 [-1,2]: f(u)>f(v) \rightarrow [u,b]=[0,5;2]. k=2: f(u)=0,062463 \leq f(v)=0,062513 \rightarrow [a,v]=[0,5;1,25]. 10$ iterasi: $[0,9980; 1,0039]. \uparrow$ maks $-(x-3)^2, [0,4], \epsilon=0,02: u=1,99, v=2,01; f(u)<f(v) \rightarrow [1,99;4]$.

6 · Fibonacci search

$F_0=F_1=1, F_n=F_{n-1}+F_{n-2}: 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987 (n=0..15)$

$$\text{pilih } n: F_n > (b-a)/l$$

($l > \epsilon$)

$$u = a + F_{n-k-1}/F_{n-k+1} \cdot (b-a); v = a + F_{n-k}/F_{n-k+1} \cdot (b-a)$$

- $\min: f(u) \leq f(v) \rightarrow b=v, v \leftarrow u$ lama, hitung u baru. Else $a=u, u \leftarrow v$ lama, hitung v baru (1 evaluasi baru)
- Iterasi $k=n: u=v \rightarrow$ pakai $v=u+\epsilon$ (itulah fungsi ϵ)

Slide: $[-1,2], l=0,01: (b-a)/l=300 \rightarrow F_{12}=233 < 300 < F_{13}=377 \rightarrow n=13. u_1=0,1459, v_1=0,8541 \rightarrow$ akhir $[0,9973; 1,0053]$

$[-2,8], l=0,1: 100 \rightarrow n=11 (F_{10}=89 < 100). b-a=10, l=0,01: 1000 \rightarrow n=16 (F_{15}=987).$

7 · Golden section

$$u = a + (1-\gamma)(b-a); v = a + \gamma(b-a)$$

$\gamma=0,618$ (golden ratio); $\gamma^2=1-\gamma \approx 0,382$; henti $b-a < l$; tanpa n , tanpa ϵ

- $\min: f(u) \leq f(v) \rightarrow a$ tetap, $b=v, v \leftarrow u$ lama, u baru $= a+(1-\gamma)(b-a)$
- $f(u) > f(v) \rightarrow a=u, b$ tetap, $u \leftarrow v$ lama, v baru $= a+\gamma(b-a)$
- Titik lama dipakai ulang → **1 evaluasi baru/iterasi**

Slide: $[-1,2]: u_1=0,1460, v_1=0,8540, f=0,7293 > 0,0213 \rightarrow a=0,146, u_2=0,8542 (=v_1), v_2=1,2918. 13$ iterasi: $[0,9969; 1,0062]$

$[0,5]: u=0,382 \times 5=1,910; v=0,618 \times 5=3,090.$

8 · Bisection (turunan)

$$m = (a+b)/2$$

henti jika $|f'(m)| < \epsilon$; f konveks & dapat diturunkan; tanpa nilai f

Contoh slide: $\min (x-1)^2, [-1,2], \epsilon=0,001, f'=2(x-1)$

k	a	b	m	f'(m)
1	-1	2	0,5	-1,0
2	0,5	2	1,25	0,5
3	0,5	1,25	0,875	-0,25

12 iterasi → $[0,9995; 1,0010]$. Panjang setelah k pembagian $= (b-a)/2^k; \uparrow [-1,2], \leq 0,01: 3/2^k \leq 0,01 \rightarrow k=9. \uparrow x^4-4x$ pada $[0,3]: [0;1,5], [0,75;1,5], [0,75;1,125]$.

9 · Aturan arah update (min vs maks)

	Min	Maks
buang kanan → $[a,v]$	$f(u) \leq f(v)$	$f(u) \geq f(v)$
buang kiri → $[u,b]$	$f(u) > f(v)$	$f(u) < f(v)$
bisection → $[a,m]$	$f'(m) > 0$	$f'(m) < 0$
bisection → $[m,b]$	$f'(m) < 0$	$f'(m) > 0$
direct search: lanjut $d=+1$	$f(x+\lambda) < f(x)$	$f(x+\lambda) > f(x)$
direct search: balik $d=-1$	$f(x+\lambda) > f(x)$	$f(x+\lambda) < f(x)$

$f'(m)=0 \rightarrow$ berhenti. Min: $f'>0$ berarti f naik di $m \rightarrow$ minimum di kiri. Maks: $f'>0$ berarti masih naik → maksimum di kanan: $-(x-1)^2$ pada $[-1,2], m=0,5, f'=1 \rightarrow [0,5;2].$ Min pada $[0,4], m=2, f'=3>0 \rightarrow [0;2].$